INTELLIINSPECT AI

Industrial Vision & Automated QA System

CERTIFICATE: REP-20261031-0001
DATE: 2026-10-31 22:42:55 UTC
STANDARD: ISO 9001:2015 / ASTM E155

INSPECTION DECISION: ACCEPT

Status: OK | Severity: NONE | Confidence: 73.0%

Product Code:	CAST-26-X101	**Machine ID:**	DC-CELL-01
Component Type:	Pump Impeller (Cast Iron/Alloy)	**Batch Number:**	B2026-01
Inspection Model:	PatchCore ResNet-18 (512x512)	**Inspector:**	IntelliInspect AI Vision Station #1

Visual Defect Localization & Anomaly Map

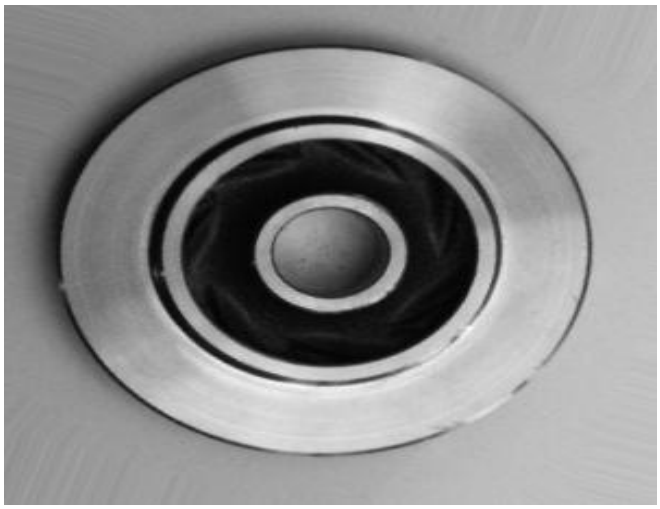


Figure 1: As-Received Casting Surface

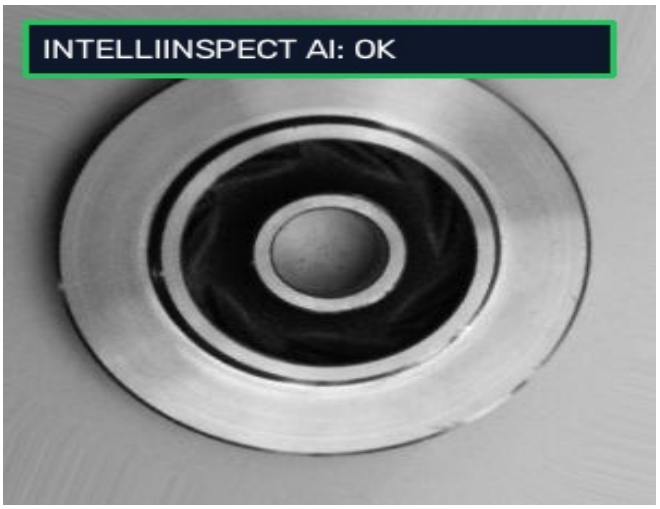


Figure 2: AI Thermal Anomaly Map & Bounding Box

Quantitative AI Measurements

Parameter	Measured Value	Standard Limit	Compliance
Anomaly Peak Score	1.5354	Threshold: 1.6577	PASS
Model Confidence	73.0%	>= 70.0%	PASS
Defect Classification	None (Defect-Free)	None (Defect-Free)	PASS
Severity Rating	NONE	NONE / LOW (Acceptable)	PASS

Smart Decision Explanation & Engineering Rationale

Engineering Notes: Component surface meets ISO 9001 casting purity tolerance. No anomalous porosity, shrinkage, or crack detected.

Digital Verification:

Scan QR code with any mobile device to verify certificate authenticity on the blockchain/cloud traceability registry.



Quality Assurance Sign-Off:
Lead QA Inspector: *A. Sharma, PE*
Station: *Die-Cast Cell #2*
Status: **VERIFIED RECORD**